

GSMA Mobile Money API Specification & Interoperability Guidelines

P2P Transfers, Merchant Payments, Cash-In/Cash-Out & Asynchronous Callback Architecture

Standard Setter:	GSM Association (GSMA Mobile Money Programme)
Specification ID:	GSMA-MM-API-V2
Classification:	Telecom & Mobile Financial Services Standard
Verification:	Integrated in Ariya Sarrafzadeh Production Architecture

1. API Architecture & Domain Model

Harmonized REST API specifications enabling mobile money operators, banks, and merchant platforms to interconnect for digital payments across emerging markets.

2. Asynchronous Polling & Callback Webhooks

Transactions decouple request ingestion from execution. Systems return HTTP 202 Accepted with a correlation token, followed by cryptographically signed asynchronous webhooks once network settlement completes.

3. Idempotency & HMAC Security Protocol

Requests enforce X-Correlation-ID and X-Idempotency-Key headers, combined with HMAC-SHA256 signature verification over request payloads to ensure non-repudiation.

4. Agent Liquidity & Sub-Wallet Structures

Hierarchical wallet architectures managing master pools, agent floats, escrow balances, and customer sub-wallets with real-time balance reservation state machines.

Systems Verification Notice: This authoritative specification establishes the invariant compliance baseline for production fintech switches, settlement reconciliation engines, and double-entry ledger implementations documented across the portfolio.